

# RELEASE\_TORRENT\_UPLOAD\_FIX

## Release Report: Torrent File Upload Fix

**Revision:** 1 **Last modified:** 2026-06-06T00:00:00Z

**Date:** 2026-04-03

**Severity:** Critical (complete loss of .torrent file upload functionality)

**Status:** Fixed, tested, deployed

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### Problem Description

Users were unable to download any torrent opened from a .torrent file through the qBittorrent WebUI. When uploading a .torrent file, setting a destination path, and submitting, the download never appeared in the downloads list and never started. Magnet links and search plugin downloads continued to work.

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### Root Cause

The download\_proxy.py proxy server (sitting in front of qBittorrent on port 7186) attempted to decode **all** POST request bodies to /api/v2/torrents/add as UTF-8 text at line 101:

```
body_str = body.decode("utf-8")
```

When a .torrent file is uploaded via the WebUI, the browser sends it as multipart/form-data containing **raw binary data** (the bencoded torrent file). This binary content is not valid UTF-8, causing a UnicodeDecodeError:

'utf-8' codec can't decode byte 0x81 in position 530: invalid start byte

The exception was caught by the generic except Exception handler at line 131, which returned an HTTP 500 error silently. Users saw no error message in the WebUI - the upload simply disappeared.

## Why magnet links worked

Magnet URLs are sent as application/x-www-form-urlencoded POST data (plain text), which IS valid UTF-8. The proxy only intercepted these for RuTracker URLs; all other magnets passed through correctly.

## Why search plugin downloads worked

Search plugin downloads go through a different code path (nova2dl.py) that the proxy specifically handles for RuTracker URLs.

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## Architecture Context

```
User Browser
  |
  v
:7186 download_proxy.py (Python HTTP proxy)
  |      - Intercepts RuTracker URLs -> downloads via
nova2dl.py
  |      - Passes through all other requests
  v
:7185 qBittorrent WebUI (native)
```

The proxy exists to handle RuTracker authentication for torrent downloads. It intercepts all API requests but should pass through non-RuTracker traffic unchanged.

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## Fix Applied

**File:** plugins/download\_proxy.py

## Changes

### 1. Added multipart detection method

`_is_multipart_file_upload()` that checks the Content-Type header for multipart/form-data.

2. **Added early bypass for file uploads:** When a multipart file upload is detected, the proxy immediately passes the request through to qBittorrent without attempting to decode the body.
3. **Added binary body fallback:** Even if Content-Type check is bypassed, a try/except `UnicodeDecodeError` around the decode call ensures binary data doesn't crash the proxy.

## Code diff (handle\_request method)

### Before:

```
def handle_request(self, body):
    try:
        path = urllib.parse.urlparse(self.path).path
        if path == "/api/v2/torrents/add" and self.command ==
            "POST" and body:
            body_str = body.decode("utf-8") # CRASHES on binary
            torrent data
            params = urllib.parse.parse_qs(body_str)
            urls = params.get("urls", [""])[0]
            ...
```

### After:

```
def _is_multipart_file_upload(self):
    content_type = self.headers.get("Content-Type", "")
    return "multipart/form-data" in content_type

def handle_request(self, body):
    try:
        path = urllib.parse.urlparse(self.path).path
        if path == "/api/v2/torrents/add" and self.command ==
            "POST" and body:
            if self._is_multipart_file_upload():
                logger.info("Multipart file upload detected,
                    passing through directly")
                self.proxy_to_qbittorrent(body)
                return

            try:
                body_str = body.decode("utf-8")
            except (UnicodeDecodeError, ValueError):

                logger.info("Binary body detected, passing through
                    directly")
                self.proxy_to_qbittorrent(body)
                return

            params = urllib.parse.parse_qs(body_str)
            urls = params.get("urls", [""])[0]
            ...
```

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# Test Suite

**File:** tests/test\_torrent\_file\_upload.py

**Tests:** 19 tests across 3 test classes

## Test Torrent Files

Downloaded 4 Linux distribution torrent files with high seeder counts for testing:

| File                                   | Size   | Source              |
|----------------------------------------|--------|---------------------|
| ubuntu-24.04-desktop-amd64.iso.torrent | 474 KB | releases.ubuntu.com |
| ubuntu-20.04-desktop-amd64.iso.torrent | 325 KB | releases.ubuntu.com |
| ubuntuserver.torrent                   | 247 KB | releases.ubuntu.com |
| debian.iso.torrent                     | 50 KB  | cdimage.debian.org  |

Located in: tests/test\_torrents/

## Test Classes

### TestProxyInfrastructure (5 tests)

- qBittorrent direct accessibility (port 7185)
- Proxy accessibility (port 7186)
- Version match through proxy
- Transfer info passthrough
- Local torrent file availability

### TestTorrentFileUpload (10 tests)

- Direct upload to qBittorrent (bypass proxy) - baseline
- Proxy upload returns "Ok." (not 500)
- Uploaded torrent appears in qBittorrent list
- Uploaded torrent has valid download state
- Magnet URL passthrough through proxy
- **All 4 distro torrent files upload successfully through proxy**
- Upload with custom save path
- Duplicate torrent detection through proxy
- **Regression: proxy never returns 500 for multipart uploads (5 iterations)**
- Preferences API passthrough through proxy

## TestProxyRegression (4 tests)

- No UnicodeDecodeError on any torrent file (tests all 4)
- Binary torrent with all byte values (0-255) doesn't crash proxy
- Proxy logs contain no errors after uploads
- Largest torrent file (474 KB) uploads without issues

## Test Results

Ran 19 tests in 47.951s - OK

ALL TORRENT FILE UPLOAD TESTS PASSED

The download proxy correctly handles .torrent file uploads.

Tested with 4 distro torrent file(s).

Tests verified on freshly restarted containers (full stop/start cycle).

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## Files Changed

| File                                  | Change                                          |
|---------------------------------------|-------------------------------------------------|
| plugins/download_proxy.py             | Fixed multipart file upload handling            |
| tests/<br>test_torrent_file_upload.py | New comprehensive test suite (19 tests)         |
| tests/test_torrents/                  | New directory with 4 Linux distro torrent files |

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## Deployment

Containers restarted and verified: - qbittorrent - qBittorrent v5.1.4-r2-ls444 on port 7185 - qbittorrent-proxy - Python proxy on port 7186

## Manual Testing

The WebUI is now available at: **http://localhost:7186**  
Credentials: admin / admin

To manually verify: 1. Open http://localhost:7186 in browser 2. Login with admin/admin 3. Click the "+" icon or drag a .torrent file onto the window 4. Set a destination path (e.g., /downloads/) 5. Click "Add" - the torrent should now appear and start downloading

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## Running Tests

```
# Run the torrent file upload test suite  
python3 tests/test_torrent_file_upload.py
```

```
# Run all existing test suites  
./run-all-tests.sh
```

```
# Quick validation  
./test.sh
```